

Client-Server Architecture

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What is Client-Server Architecture?

A model where clients request resources and servers provide them.

Used in web applications, email, file sharing, and more.

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Main Components

Client: Sends request, waits for and uses the response.

Server: Receives request, processes it, and sends back response.

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How It Works

Client sends a request to the server.

Server processes the request.

Server sends a response back to the client.

Client uses the returned data or service.

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Visual Representation (Text)

[Client] ---> [Request] ---> [Server]

[Client] <--- [Response] <--- [Server]

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Examples

Web browsing: Browser (Client) and Website (Server)

Email: Email App (Client) and Mail Server

Online Shopping: App/Browser and E- commerce Server

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Advantages

Centralized control and data management.

Easier to secure and monitor.

Scalable and easier to maintain.

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Disadvantages

Single point of failure: if server is down, clients are affected.

Server may become overloaded with too many requests.

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